

FluxDash Software Documentation

Version 1.0.0 | Teaching-Oriented Eddy Covariance Workflow App

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FluxDash was created to help students and early-career professionals learn how to analyze eddy covariance flux datasets in a single guided environment.

The software puts viewing, preprocessing, optional EddyPro processing, REddyProc gap filling and partitioning, footprint diagnostics, unit conversion, result visualization, and workflow documentation under one roof.

Intended users

- Students learning eddy covariance data processing, flux partitioning, and interpretation.
- Early-career professionals who need a structured introduction to flux analysis workflows.
- Instructors who want a lab-friendly interface without making students manage every processing tool manually.
- Researchers who want quick exploratory processing before validating final publication-grade outputs.

1. What FluxDash Does

FluxDash is a workflow-integration and teaching application for eddy covariance data. It does not hide the scientific decisions. Instead, it exposes the key choices in a structured form so users can understand how each decision affects the final NEE, GPP, Reco, ET, gap-filling, and footprint outputs.

Stage	Purpose	Main outputs
View	Scan a data folder, load EP-Summary or processed flux outputs, inspect variables and date ranges.	Time series, scatter plots, quick KPIs, variable lists.
Process	Extract SmartFlux/EddyPro .ghg outputs, optionally run EddyPro, or use raw covariance fallback for older .ghg archives.	fluxdash_processed.csv with flux, driver, QC, footprint, and prefilter flags.
Pre-filter	Convert missing-value codes and unrealistic values to NA before gap filling.	Filtered NEE, LE, H, Rg/SW plus flag columns.
Partition	Run REddyProc for u^* filtering, MDS gap filling, and NEE partitioning.	Gap-filled NEE, GPP, Reco, u^* threshold, gap diagnostics.
Results	Explore budgets, cumulative fluxes, response curves, footprint plots, and method diagnostics.	Downloadable charts, tables, and manuscript-ready workflow logs.
Save/Reload	Save the working state and reopen a project later.	.fluxdash.json project file.

2. Installation and Required Environments

- Windows installer: distribute the built FluxDash Setup 1.0.0.exe to students.
- Python 3: required for the built-in .ghg extraction and raw covariance fallback.
- R: required for the Partition tab.
- REddyProc R package: used for u^* filtering, MDS gap filling, and flux partitioning.
- EddyPro: optional. Use it only when site metadata and EddyPro file templates are configured correctly.

3. Data Types Supported

Input type	How FluxDash uses it	Notes
EP-Summary / processed CSV or TXT	Loaded directly for View and can be used as a source for Partition.	Fastest route when EddyPro outputs already exist.
SmartFlux/EddyPro .ghg with embedded output	Built-in extractor reads embedded full_output and biomet files.	Good for newer SmartFlux files that already contain processed records.
Older .ghg raw archives	Raw covariance fallback reads high-frequency .data and biomet records.	Teaching/exploratory fallback; validate against EddyPro for publication.
Biomet .data files	Used to merge radiation, temperature, RH/VPD, PPFD/PAR, and other drivers.	Needed for reliable gap filling and response analyses.

4. Process Tab

The Process tab converts raw or embedded .ghg information into a half-hourly flux table that can be handed to REddyProc. The default recommendation for class use is the built-in .ghg processor/extractor, because it avoids EddyPro file-template mismatch errors and works with SmartFlux files that already contain usable flux outputs.

Option	Meaning	Manuscript/log relevance
Engine to use	Built-in .ghg processor/extractor, EddyPro, or fallback behavior.	State whether outputs came from EddyPro, SmartFlux embedded output, or raw covariance fallback.
Averaging period	Usually 30 minutes for EC flux processing.	Needed for unit conversion from flux rates to period sums.
Site metadata	Latitude, longitude, altitude, measurement height, canopy height, roughness length.	Supports partitioning, footprint interpretation, and reproducibility.
Corrections	WPL, spectral corrections, footprint estimation, tilt correction, QC scheme.	Report correction and QC choices in methods.
Output subfolder	Where FluxDash writes fluxdash_processed.csv.	Stored in workflow log and project file.

5. Pre-filtering Before Gap Filling

Pre-filtering is applied before REddyProc gap filling so unrealistic spikes or missing-value codes do not enter the gap-filling and partitioning model. FluxDash keeps the timestamps and converts rejected values to NA by default so the half-hourly time grid remains regular.

Default filter	Default range	Reason
NEE / CO2 flux	-50 to +50 $\mu\text{mol m}^{-2} \text{s}^{-1}$	Removes extreme CO2 flux spikes before gap filling.
Latent heat flux (LE)	0 to 700 W m^{-2}	Removes physically unrealistic ET/LE values for most teaching datasets.
Sensible heat flux (H)	-100 to 700 W m^{-2}	Allows negative sensible heat while excluding large spikes.
Radiation Rg/SW	0 to 1600 W m^{-2}	Flags negative radiation and extreme shortwave values when enabled.
Missing-value codes	-9999, -99999	Converted to NA before analysis.

Users may edit these ranges based on site type, instrument behavior, climate, and instructor guidance. The workflow log records the exact limits used.

6. Partition Tab: Gap Filling and Flux Partitioning

The Partition tab uses REddyProc to filter low-turbulence periods, gap-fill NEE, and partition NEE into GPP and ecosystem respiration. This is the core step that turns half-hourly flux data into interpretable carbon budgets.

Choice	What it controls	Recommended reporting
Partitioning method	Nighttime/Reichstein, daytime/Lasslop, or both.	State the method and whether final GPP/Reco came from nighttime, daytime, or comparison outputs.
u* mode	Auto-estimated bootstrapped threshold or manual threshold.	Report u* threshold, confidence interval when available, and percentage removed.
MDS gap filling	REddyProc gap-fills missing NEE using meteorological drivers.	Report method, software, gap fraction, and uncertainty where available.
Latitude/longitude/time zone	Used by REddyProc to define radiation/day-night context.	Confirm site metadata and time zone in methods.
Input CSV	Usually the latest fluxdash_processed.csv.	Recorded in the workflow log.

7. Results Tab

The Results tab is designed as an interactive analysis explorer. Users select a preset, period range, grouping interval, statistic, unit conversion, and metric set. Charts and tables can be downloaded for reports or lab assignments.

Preset group	Examples	Use
Carbon budgets	Cumulative NEE/GPP/Reco, source/sink, NEP, CUE.	Annual/monthly/daily budgets and source-sink interpretation.
Energy and water	LE, H, ET, Bowen ratio, EBC, WUE.	Water-cycle and energy-balance interpretation.
Diagnostics	Gap filling, u* threshold, method comparison.	Evaluate data quality and processing effects.
Driver responses	Light response, VPD response, Q10 temperature response.	Mechanistic interpretation for manuscripts and lab reports.
Footprint	Kljun/EddyPro contour climatology and source-distance point view.	Assess site representativeness and fetch direction.

8. Light-Response and Q10 Plots

FluxDash v1.0 allows users to plot NEE or GPP against SW/Rg, PAR, or PPFD. Users can show raw points, binned means, fitted curves, and fitted equations. For NEE light response, the displayed model is rectangular-hyperbola style:

$$NEE = R_d - (GPP_{max} \times I) / (K + I)$$

For GPP light saturation, the displayed model is:

$$GPP = (GPP_{max} \times I) / (K + I)$$

For temperature sensitivity, FluxDash fits the log-linear form $Reco = \exp(a + bT)$ and reports $Q10 = \exp(10b)$. These fits are intended for teaching and screening; publication-quality analyses should be reviewed against accepted EC modeling practice and site-specific assumptions.

9. Analysis-Level Outlier Handling

The Results tab includes outlier tools for response plots. These filters affect the visualization and exported analysis table, not the underlying processed or partitioned file. They are useful when a response plot contains obvious driver artifacts, such as repeated radiation values that create a vertical cluster.

Outlier option	How it works	When to use
No analysis outlier filter	Plots all finite selected points.	Default for transparency.
Remove IQR outliers	Removes points outside $1.5 \times \text{IQR}$ limits in x/y.	General exploratory cleanup.
Trim x/y percentiles	Removes extreme tails based on selected percent.	When a few extremes dominate the axis.
Manual x/y range	Keeps only points within user-defined x and y ranges.	When the instructor knows valid driver/flux bounds.
Remove repeated x-spike clusters	Removes heavily repeated x values using the selected tolerance.	For suspicious vertical bands around one radiation/driver value.

10. Footprint Visualization

FluxDash can visualize footprint source-distance outputs and Kljun/EddyPro percentile columns when they exist in the processed data. The contour explorer aggregates footprint percentile distances by wind direction and can overlay contours on an optional background image. This is a diagnostic visualization, not a full GIS georeferencing workflow.

11. Save/Reload and Workflow Log

Use Save project to create a .fluxdash.json file that stores folder paths, settings, selected variables, processing outputs, partitioning results, result controls, footprint background, and workflow events. Use Open project to continue later.

The workflow log is designed for reproducibility. It records the user choices from processing through partitioning and results, including references supporting the selected methods. It can be exported as a text file and used when drafting a methods section or reviewing a student lab submission.

12. Help Menu

- Help > About FluxDash opens the software purpose, author, copyright, and reference list.
- Help > Open Software Documentation (PDF) opens the bundled PDF copy of this document.
- Help > Open Software Documentation (Word) opens the editable DOCX copy of this document.

13. Sign Conventions and Unit Notes

- NEE follows the atmospheric convention: negative NEE means net ecosystem CO₂ uptake/sink; positive NEE means net ecosystem CO₂ release/source.
- GPP and Reco are shown as positive magnitudes in most FluxDash result plots.
- NEP is calculated as -NEE, so positive NEP indicates carbon gain from the ecosystem perspective.
- Flux-rate to carbon-sum conversions use the selected time step, typically 1800 seconds for 30-minute data.
- LE-to-ET conversion uses latent heat of vaporization approximated as 2.45 MJ kg⁻¹.

14. Known Limitations and Best Practices

- Do not treat the raw covariance fallback as a complete EddyPro replacement for manuscripts.
- Always review input units, time zone, measurement height, radiation drivers, VPD/Tair availability, and gap fractions.
- Use pre-filtering carefully. Removing too much data can bias gap filling and annual budgets.
- Outlier filters in the Results tab are for analysis visualization and table export only; they do not change the original partitioned CSV.
- For publication, document instrument setup, correction methods, QC criteria, u^* threshold method/value, gap-filling method, partitioning method, footprint model, annual budget uncertainty, and sign convention.

15. Reference Library

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FLUXNET/AmeriFlux CSV Dataset Support

FluxDash v1.0 can now operate on standardized FLUXNET and AmeriFlux CSV datasets in addition to .ghg and EP-Summary files. This allows instructors to use public flux products for teaching when raw tower files are not available or when the goal is to focus on interpretation rather than raw high-frequency processing.

Supported FLUXNET-style files

- FULLSET_HH: half-hourly full data product. Preferred for detailed response curves, diel patterns, and cumulative budgets.
- FULLSET_DD, FULLSET_WW, FULLSET_MM, FULLSET_YY: daily, weekly, monthly, and yearly products. Useful for temporal summaries and annual budget exercises.
- ERA5_HH, ERA5_DD, ERA5_WW, ERA5_MM, ERA5_YY: meteorological driver products. Useful for climate-driver comparisons when direct tower variables are limited.
- AUXNEE and AUXMETEO files: read as supporting metadata where available; the main Results Explorer uses FULLSET products first.

Column mapping used by FluxDash

FluxDash variable	Preferred FLUXNET/AmeriFlux columns	Notes
NEE / co2_flux	NEE_VUT_REF, NEE_CUT_REF, NEE_VUT_USTAR50, NEE_CUT_USTAR50	Atmospheric convention: negative values indicate ecosystem CO2 uptake.
GPP	GPP_NT_VUT_REF, GPP_DT_VUT_REF, GPP_NT/DT USTAR50 or MEAN variants	Shown as positive carbon uptake in Results.
Reco / RE	RECO_NT_VUT_REF, RECO_DT_VUT_REF, RECO_NT/DT USTAR50 or MEAN variants	Shown as positive ecosystem respiration.
LE	LE_F_MDS, LE_CORR, LE	Latent heat flux in W m ⁻² .
H	H_F_MDS, H_CORR, H	Sensible heat flux in W m ⁻² .
Radiation / Rg	SW_IN_F_MDS, SW_IN_F, SW_IN_ERA, SW_IN	Used for light-response analyses when PPFD/PAR is unavailable.
PPFD / PAR	PPFD_IN, PPFD	Preferred driver for photosynthetic light-response curves when present.
Tair	TA_F_MDS, TA_F, TA_ERA	Air temperature for Q10 and driver analyses.
VPD	VPD_F_MDS, VPD_F, VPD_ERA	Used for stomatal/VPD response plots.
u*	USTAR	Used as a turbulence diagnostic where available.

How FluxDash handles FLUXNET FULLSET data

- When a FULLSET file is detected, FluxDash normalizes its standardized columns into the internal FluxDash names used in View, Partition, and Results.
- Because FULLSET products already include gap-filled and partitioned variables, the Partition tab does not rerun REdDyProc. Instead, it uses the standardized NEE, GPP, Reco, LE, H, radiation, temperature, VPD, and QC columns directly.
- The workflow log records that FLUXNET/FULLSET products were used and includes references supporting the standardized FLUXNET/ONEFlux processing approach.

- Students can still use Results Explorer tools such as cumulative carbon budgets, source/sink classification, seasonal patterns, light response, temperature response/Q10, VPD response, ET, WUE, CUE, and energy-balance diagnostics when the required columns are present.

Teaching note

FLUXNET data are ideal for teaching because they allow students to explore real flux products without first configuring raw high-frequency processing. However, instructors should explain that the pre-processing, u^* filtering, gap filling, and partitioning have already been applied by the data-product pipeline. FluxDash therefore treats FLUXNET FULLSET files as standardized products rather than raw tower data.

Additional references for FLUXNET support

- Pastorello, G. et al. (2020). The FLUXNET2015 dataset and the ONEFlux processing pipeline for eddy covariance data. Scientific Data.
- AmeriFlux Management Project. AmeriFlux BASE and FLUXNET data-product documentation.
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